



كلية الهندسة المعلوماتية

السنة الرابعة

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نظم التشغيل

1. Introduction To Linux

Multiple Choice Questions (MCQ)

1. Which component of the Linux Operating System acts as the primary interface between the software applications and the physical hardware?

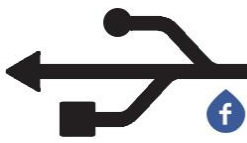
- a) The Shell
- b) The Terminal
- c) The Kernel
- d) The GUI

2. In the Linux File System Hierarchy, which directory is traditionally used to store system configuration files?

- a) /bin
- b) /etc
- c) /root
- d) /home

3. Which command is used to see a list of all previously executed commands in the current session?

- a) uptime
- b) whereis
- c) history
- d) jobs



4. If you are searching for a line in /etc/passwd that specifically *ends* with the string "sh", which grep command should you use?

- a) `grep ^sh /etc/passwd`
- b) `grep sh* /etc/passwd`
- c) `grep sh$ /etc/passwd`
- d) `grep .sh /etc/passwd`

5. If a user is currently in /home/student/docs and executes `cd ../../teacher`, what is the resulting working directory?

- a) /home/teacher
- b) / teacher
- c) /home/student/ teacher
- d) This will result in an error.

6. The command `split -l 50 data.txt` will perform which action?

- a) Delete the first 50 lines of data.txt.
- b) Search for the number 50 in data.txt.
- c) Increase the size of data.txt by 50 bytes.
- d) Break data.txt into multiple files, each containing 50 lines.

7. In the Vim editor, which key or command sequence is used to switch from "Insert Mode" back to "Command Mode" and then save and exit?

- a) Esc then `:wq`
- b) Ctrl+C then `:q!`
- c) Shift+S then exit
- d) Esc then `:q`

8. Which grep pattern would you use to find all lines that contain a word starting with 'a', followed by exactly two characters, and then the letter 'z'?

- a) `grep a*z`
- b) `grep a..z`
- c) `grep ^a.z$`
- d) `grep a...z`

9. Which statement best defines an operating system?

- a) A collection of application programs for users
- b) Software that manages hardware and provides services to applications
- c) A firmware stored permanently on hardware



d) A programming language for system development

10. Which component of Linux directly communicates with the hardware?

- a) Shell
- b) Applications
- c) Kernel
- d) Daemons

11. Which layer includes systemd, bash, and background services?

- a) Kernel space
- b) Application layer
- c) Hardware abstraction layer
- d) System user space

12. Which is an example of a daemon?

- a) Bash
- b) systemd
- c) Network logging service
- d) b and c

13. What is the primary role of a package manager such as apt?

- a) Start and stop system services
- b) Compile applications from source only
- c) Manage installation and updates of software
- d) Allocate system memory

14. Which statement about the shell is correct?

- a) It replaces the kernel
- b) It interprets commands and communicates with the kernel
- c) It directly controls hardware
- d) It is only graphical

15. What does the pipe operator | do?

- a) Redirects output to a file
- b) Combines files
- c) Sends output of one command as input to another
- d) Runs commands in parallel

16. When we say that Linux is open source, this means:

- a) The source code is available to everyone for modification and redistribution
- b) It is free for personal use only
- c) It does not require a license for commercial use
- d) It supports all types of hardware

17. What is the difference between init and systemd?

- a) init uses sequential scripts, while systemd uses parallel units
- b) init is faster, systemd is slower
- c) init is for GUI only, while systemd is for CLI only
- d) They are the same

18. Why is the Linux file system structure called hierarchical?

- a) Because it is organized according to file size
- b) Because it starts from a root (/) and branches into subdirectories
- c) Because files are arranged by age
- d) Because there are important files and secondary files

19. What is the correct relationship between the kernel and the shell?

- a) The shell is part of the kernel
- b) The kernel directly interprets user commands
- c) The shell is an interface between the user and the kernel
- d) The kernel is the graphical interface of the system

20. When we say that Linux is a multiuser system, this means:

- a) Multiple people can use the system at the same time but with the same account
- b) Each user has a separate account with their own permissions and files
- c) The system supports only 10 users as a maximum
- d) All users have root privileges by default

21. What is the output of the following command?

`touch file{1..3}.txt`

- a) Creates a file named file{1..3}.txt
- b) Creates 3 files: file1.txt, file2.txt, file3.txt
- c) The command is incorrect
- d) Creates a directory named file

22. Which command is used to delete a file with confirmation?

- a) `rm file.txt`
- b) `rm -f file.txt`
- c) `rm -i file.txt`
- d) `rm --yes file.txt`

23. What is the function of `dd` in Vim?

- a) Deletes the current line
- b) Copies the current line
- c) Pastes the line
- d) Saves the file

24. Is there an error in the following command?

```
grep "Hi" *.txt << output
```

- a) There is an issue with `Hi`
- b) There is an issue with redirection
- c) The command is correct
- d) There is an issue with the wildcard, i.e., `*.txt`

25. Which command will produce an error if the user is a regular (non-root) user?

- a) `ls ~/`
- b) `whoami`
- c) `rm /etc/passwd`
- d) `pwd`

26. Which commands produce the same output?

```
cat file.txt | grep "text"
```

```
grep "text" < file.txt
```

```
grep "text" file.txt
```

```
find "text" file.txt
```

- a) 1 and 2 only
- b) 1 and 3 only
- c) 1, 2, and 3
- d) All of them produce the same output

27. Which command shows only directories in the current directory?

- a) `ls -d */`
- b) `ls -f`
- c) `ls -a`
- d) `ls -l`

28. What is the output of the following command?

`mkdir test && cd test && pwd`

- a) `/home/user`
- b) `/test`
- c) `/home/user/test`
- d) The command is incorrect

29. What is the output of the following command?

`grep -c "error" log.txt`

- a) Displays the lines that contain "error"
- b) Counts how many lines contain "error"
- c) Deletes the lines that contain "error"
- d) Displays the first 10 lines

Answers

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
C	B	C	C	A	D	A	B	B	C	D	D	C	B	C
16	17	18	19	20	21	22	23	24	25	26	27	28	29	
A	A	B	C	B	B	C	A	B	C	C	A	C	B	

Write True or False, and correct the FALSE statements.

- 1) Unlike Windows, Linux grants root (administrator) access to all users by default to ensure ease of use.
- 2) The command `mkdir -p dir1/dir2` will fail if "dir1" does not already exist.
- 3) In the Linux terminal, the filename `Report.txt` and `report.txt` are treated as the exact same file.
- 4) The `mv` command is used only for moving files between directories
- 5) In Linux, the Shell is the software component that manages CPU and memory allocation directly.
- 6) The standard output redirection operator `>>` will overwrite the existing content of a file.
- 7) The kernel operates in user space
- 8) `sudo apt update` upgrades all installed packages to their latest versions.
- 9) The command `cd` without arguments moves the user to their home directory.
- 10) Vim starts in command mode by default
- 11) The command `jobs` lists background processes from all terminals.
- 12) We use the options with command to change the behaviour of the command
- 13) To go to the last line in a file opened with vim editor use option vim ee

Answers

- 1) False
Linux does NOT grant root access by default to reduce security risks.
- 2) False
The `-p` flag (parent) allows the command to create parent directories automatically if they don't exist
- 3) False
Linux is case-sensitive; `Report.txt` and `report.txt` are two distinct files.
- 4) False
The `mv` command is used for moving files/directories and renaming files/directories
- 5) False
The Kernel manages hardware (CPU/Memory); the Shell is the user interface that interprets commands.



6) False

>> appends to a file; > is the operator that overwrites.

7) False

The kernel operates in kernel space.

8) False

sudo apt update updates package lists only, not installed packages

9) True

10) True

11) False

It lists jobs only in the current shell.

12) True

13) False

we use the option G

Commands Output

Analyze the following Bash commands. Write the exact output that would appear in the terminal. If the command is syntactically incorrect, write "ERROR" and provide the corrected version of the command.

```
mkdir Lecture_Notes
cd Lecture_Notes
touch slide1.pdf slide2.pdf
ls | wc -l
```

Output: 2

Explanation: ls lists two files, and wc -l counts the lines of that list.

```
grep ^# /etc/services | head -n 2
```

Output:

The first two comment lines (lines starting with #) found in the /etc/services file.

Context: The folder Projects contains a sub-folder named C++
rmdir Projects

Output:

ERROR (Output: rmdir: failed to remove 'Projects': Directory not empty).

Corrected version: rm -r Projects.

rmdir only works on directories that are completely empty.

Explain what this command does:

```
find . -name "data" -exec rm {} \;
```

Searches for files or directories named “data” starting from the current directory and deletes each one found.

Explain what this command does:

```
ls >> f1 2>> f2
```

redirects the output of ls into two different files, depending on the type of output.

Standard output (>>) is redirected to f1

Standard error (2>>) is redirected to f2

we want to display the contents of test.txt:

```
cat << test.txt
```

Output:

Error, the operator << is a here-document operator and does not read from a file.

Correct version:

```
cat < test.txt
```

We want to get the lines that starts with rbc:

```
ls /etc/passwd | grep ^rbc
```

Output:

Error (the command ls only prints the file name and does not display its contents.)

**Correct version:**

```
cat /etc/passwd | grep ^rbcs
```

- When ls is given a file that exists, it prints the file name or path exactly as it was passed to it.
- When the file does not exist, it produces an error message.
- In both cases, ls never prints the content of the file; it only works on file names and directories.

2. Users and Groups in Linux

Multiple Choice Questions (MCQ)

1. Who is the user that has full system access in Linux?

- a) Regular User
- b) Admin
- c) Root

2. What type of account is dedicated to services such as mysql and www-data?

- a) Power User
- b) Human User
- c) Super User
- d) System User

3. In which file is the Default Shell path for each user stored?

- a) /etc/shadow
- b) /etc/passwd
- c) /etc/group
- d) /bin/bash

4. What command is used to create a new user interactively in Ubuntu?

- a) useradd -interactive
- b) newuser
- c) createuser
- d) adduser

5. Which directory contains the default configuration files (such as .bashrc) that are copied for each new user?

- a) /etc/skel
- b) /home/template
- c) /etc/defaults
- d) /root/skel

6. Where are basic account information such as the username and UID stored?

- a) /etc/shadow
- b) /home
- c) /etc/passwd
- d) /etc/group

7. Which field in the /etc/passwd file specifies the Login Shell?

- a) The sixth field
- b) The seventh and last field
- c) The fourth field
- d) The first field

8. Which command is used to change a password?

- a) Passwd
- b) Chown
- c) usermod

9. In the output of ls -l, what does the first character represent?

- a) The type of the item (directory or file)
- b) The owner name
- c) The file size
- d) The modification date

10. What is the numeric value of the Read permission?

- a) 2
- b) 1
- c) 4
- d) 7

11. To change the owner, we use the command:

- a) Chmod

- b) Chgrp
- c) Chown
- d) usermod

12. What does the command `chmod +x file` do?

- a) Adds execute permission for the owner only
- b) Adds execute permission for everyone
- c) Adds read permission
- d) Changes the owner

13. What is the UID of the root user?

- a) 0
- b) 500
- c) 1
- d) 1000

14. The `-s` option in `usermod` changes:

- a) The password
- b) Makes the account secure
- c) Changes the shell
- d) The home directory

15. What does the symbol `d` at the beginning of permissions represent?

- a) Document
- b) Deleted
- c) Directory
- d) Data

16. When using the command
`useradd -e 2025-12-31 username`

What is the purpose of the `-e` option?

- a) Set the home directory (Entry directory)
- b) Set the account expiration date (Expiry Date)
- c) Set the execution shell
- d) Encrypt the password

17. Why might we use the -f option with the userdel command?

- a) To force the system to delete the user even if they are currently logged in
- b) To delete the user's home directory
- c) To change the username quickly (Fast rename)
- d) To delete only the user's groups

18. Which file contains the actual list of groups and their numeric IDs (GID)?

- a) /etc/groups.conf
- b) /etc/gshadow
- c) /etc/passwd
- d) /etc/group

19. Which of the following commands adds the user bob to an additional group called developers without removing him from his current groups?

- a) usermod -G developers bob
- b) usermod -aG developers bob
- c) chown bob:developers file
- d) useradd -g developers bob

20. What is the UID range reserved for System Users in modern Linux distributions?

- a) Only 0
- b) There is no specific range, numbers are random
- c) Always 1000 and above
- d) Usually between 1 and 999

21. What is the result of the command `chmod 000 secret.txt` for the root user?

- a) Root will still be able to read and modify the file despite the 000 permission
- b) The system will ask Root for the original owner's password
- c) Root will never be able to read the file
- d) The file will be deleted automatically

The root user bypasses standard permission checks and has full access to all files regardless of permission bits

22. Which permission allows entering a directory?

- a) r
- b) w
- c) x

d) s

23. Which command temporarily switches to another group?

- a) Su
- b) newgrp
- c) Groupmod
- d) gpasswd

24. Which is a correct symbolic permission command?

- a) chmod rwx file
- b) chmod u+rwx file
- c) chmod 777x file
- d) chmod user+r file

25. In a standard Ubuntu system, how do you grant a regular user root-level privileges (the ability to use sudo)?

- a) Add the user to the sudo group.
- b) Change the user's UID to 0 in the /etc/passwd file.
- c) Move the user's home directory to /root.
- d) Change the ownership of the /bin/bash shell to that user.

Answers

1	2	3	4	5	6	7	8	9	10	11	12	13
C	D	B	D	A	C	B	A	A	C	C	B	A
14	15	16	17	18	19	20	21	22	23	24	25	
C	C	B	A	D	B	D	A	C	B	B	A	

Write True or False, and correct the FALSE statements.

- 1) System users are mainly created for running services and daemons.
- 2) UID is always the same as GID for every user.
- 3) The command `sudo userdel username` removes the user and their home directory by default.
- 4) In permissions, x means execute for files and enter for directories.
- 5) `chmod 000 file` means no one has any permissions.
- 6) The command `chmod a+r file` gives read permission to everyone.
- 7) In `/etc/passwd`, the field marked x stores the actual password.
- 8) The root user has a UID of 1 and possesses full system access.
- 9) The command `chmod +r file.txt` is functionally equivalent to `chmod a+r file.txt`.
- 10) The primary group of a user is the group to which that user is assigned by default when they create a new file.
- 11) To change only the group ownership of a file using `chown`, the syntax is `sudo chown groupname filename`
- 12) If a file has permissions 644, the owner has the right to execute the file as a script.
- 13) Deleting a group using `groupdel` will automatically delete all users who were members of that group.

Answers

- 1) TRUE
- 2) False
UID identifies users, GID identifies groups and they can be different.
- 3) False
It deletes only the user account. Use `-r` to delete home directory.
- 4) TRUE
- 5) TRUE
- 6) TRUE
- 7) False
The actual password is stored in `/etc/shadow`.



8) False.

The root user has a UID of 0

9) TRUE

10) TRUE

11) False.

To change only the group with chown, you must use a colon:

`sudo chown :groupname filename.`

Or use `chgrp`

12) False.

Permissions 644 translate to `rw-r--r--`. The owner has read and write access, but no execute (x) access.

13) False.

`groupdel` removes the group definition, but it does not delete the users who belonged to it. Those users simply lose that group association.

Commands Output

Analyze the following Bash commands. Write the exact output that would appear in the terminal. If the command is syntactically incorrect, write "ERROR" and provide the corrected version of the command.

```
chmod 781 script.sh
```

Output:

ERROR.

Correction:

The second digit should be any number from 0 to 7.

(Assume the file 'report.pdf' exists and 'hr_dept' is a valid group)

What does this command do:

```
sudo chmod :hr_dept report.pdf
```

Output:

Error, it will give: `chmod: invalid mode ":g1"`

Correction:

```
sudo chown :hr_dept report.pdf
```

The command changes the group ownership of the file to 'hr_dept' while keeping the owner unchanged.

(Assume 'secret_key.pem' initially had permissions -rwxrwxrwx. Write the resulting symbolic permission string as seen in `ls -l`)

```
chmod u=rw , g=r , o-rwx secret_key.pem
```

the command will give error because of the spaces around the comma (,)

corrected version:

```
chmod u=rw,g=r,o-rwx secret_key.pem (no spaces before or after “,”)
```

Output: -rw-r-----

(Assume 'bob' was already a member of the 'marketing' group. Describe the outcome of the command regarding 'marketing')

```
usermod -G sudo,developers bob
```

Bob is now a member of 'sudo' and 'developers' (and he stays a member of his primary group which name is by default bob), but he has been removed from the 'marketing' group.

```
Cat /etc/groups | grep ^rbcs
```

Output:

Error:

- cat must be in lower case or it will give command not found error (linux is case-sensitive)
- /etc/groups: No such file or directory (group should be without an s)

Correction:

```
cat /etc/group | grep ^rbcs
```

Output:

All groups whose names start with 'rbcs'

3. Multitasking and Processes

Multiple Choice Questions (MCQ)

1. When a Linux computer starts, what is the very first component that runs (during boot)?

- a) Kernel
- b) BIOS
- c) systemd
- d) Bootloader (GRUB)

2. After the Kernel is loaded, what is the first process started by the Kernel?

- a) login process
- b) bash shell
- c) init / systemd
- d) Desktop Environment

3. When Linux boots, which process gets PID = 1?

- a) Kernel
- b) Shell
- c) Init
- d) daemon

4. Which tool is best for monitoring CPU usage changes over time?

- a) ps
- b) top
- c) kill
- d) init

5. What is the most accurate description of state T in ps?

- a) Process waiting for I/O
- b) Process stopped by a signal
- c) Process sleeping
- d) Process terminated

6. Which statement is NOT true about the Z (Zombie) state?

- a) The process has terminated
- b) It does not use CPU
- c) It still has an entry in the process table



d) It can execute instructions

7. Why may a process remain in Zombie state?

- a) Because it is waiting for I/O
- b) Because it has not been cleaned up by its parent yet
- c) Because it is sleeping
- d) Because it runs as a daemon

8. In STAT, what does the letter S mean?

- a) Sleep
- b) Stop
- c) Suspend
- d) Shutdown

9. If a process appears with state S in top, what can be concluded?

- a) It is using CPU
- b) It is waiting for an event
- c) It has terminated
- d) It is stopped by a signal

10. In the STAT column in ps or top, what does state R mean?

- a) The process is running
- b) The process is stopped
- c) The process is ready to run
- d) Both a and c

11. Which STAT state is directly related to debugging tools like gdb?

- a) R
- b) S
- c) t
- d) Z

12. In the STAT column in ps/top, what does state T mean?

- a) Terminate
- b) Track
- c) Stopped
- d) Sleep

13. Which option shows system processes even if they are not attached to a terminal?

- a) u
- b) a
- c) x
- d) u + a

14. In ps output, which column shows the actual resident memory in RAM?

- a) VSZ
- b) %MEM
- c) RSS
- d) TIME

15. In ps output, which column may be empty for some processes?

- a) USER
- b) PID
- c) TTY
- d) COMMAND

16. In ps columns, which shows total CPU time used since the process started?

- a) %CPU
- b) TIME
- c) START
- d) RSS

17. Which statement is NOT true about the id field in top?

- a) It represents CPU idle time
- b) A high value means CPU is not busy
- c) It indicates high CPU usage
- d) It decreases when load increases

18. In top, which field shows CPU time used by kernel processes?

- a) us
- b) ni
- c) sy
- d) wa

19. A high wa value in top usually indicates:

- a) CPU fully busy
- b) Waiting for I/O
- c) High memory usage
- d) Many user processes

20. A system with high id and large buff/cache usually means:

- a) System under heavy pressure
- b) CPU problem
- c) Normal performance
- d) Memory leak

21. Which field does NOT represent interrupt handling time in top?

- a) hi
- b) si
- c) st
- d) a and b together

22. In the top header, which field shows system uptime since last boot?

- a) Time
- b) Users
- c) Load Average
- d) Uptime

23. When memory pressure increases in Linux, what action may the system take?

- a) Use free memory to store process data
- b) Move inactive pages from RAM to swap
- c) Immediately stop processes
- d) Ignore new memory requests

24. In the process table of top, which column shows the scheduler priority?

- a) NI
- b) PR
- c) S
- d) SHR

25. A process with high NI and high PR usually means:

- a) High priority
- b) Low priority
- c) Stopped process
- d) Process using swap

26. Which statement about PR in Linux is NOT accurate?

- a) It is calculated internally by the system
- b) It is affected by the nice value
- c) The user can set it directly
- d) It can change during execution

27. Which process is most likely to run first?

- a) Process A: PR=20, NI=10
- b) Process B: PR=20, NI=5
- c) Process C: PR=20, NI=0
- d) Process D: PR=20, NI=-5

28. What is the possible range of PR in top?

- a) -20 to +19
- b) 0 to 99
- c) 0 to 139
- d) 1 to 139

29. Which command changes the nice value of a running process?

- a) nice
- b) top
- c) renice
- d) ps

30. A normal user runs:

`nice -n -5 manarprocess`

What is the expected result?

- a) The priority of manarprocess increases
- b) The program runs with default nice value (ni=0)
- c) The requested priority is not granted
- d) The priority of manarprocess decreases

Only root users can set a negative nice value

31. When using the renice command to change the priority of a running process, which elements must be specified?

- a) nice
- b) COMMAND + nice
- c) nice + PID
- d) COMMAND + PID + nice

32. What happens when executing the following command without specifying a signal type?

kill 2468

- a) The command is rejected because no signal is specified
- b) The process is immediately killed
- c) A graceful termination signal is sent
- d) No effect occurs

33. Which signal number is used to immediately kill a process and cannot be caught or ignored?

- a) 3
- b) 9
- c) 15
- d) 19

34. Which signal pair (name – number) is correct?

- a) SIGINT – 2
- b) SIGSTOP – 18
- c) SIGTERM – 17
- d) SIGCHLD – 19

35. Sending signal number 2 to a running process usually causes:

- a) Uncatchable termination
- b) The signal to be ignored
- c) User interrupt (Ctrl+C)
- d) Temporary stop of execution

36. Which signal may terminate a process and produce a core dump?

- a) SIGINT
- b) SIGQUIT
- c) SIGTERM
- d) SIGKILL

37. When a child process terminates, which signal number is sent to the parent?

- a) 9
- b) 15
- c) 17
- d) 19

38. Which signal has number 19?

- a) SIGSTOP
- b) SIGCONT
- c) SIGCHLD
- d) SIGTERM

39. A system administrator runs:

```
renice -n -50 -p 1111
```

What is the expected result?

- a) The process gets nice value -50 because root can exceed the limits
- b) The system automatically sets nice to -20 and succeeds
- c) The command silently fails and keeps the old nice value
- d) The command shows an error and rejects the value as out of range

Answers

1	2	3	4	5	6	7	8	9	10	11	12	13
B	C	C	B	B	D	B	A	B	D	C	C	C
14	15	16	17	18	19	20	21	22	23	24	25	26
C	C	B	C	C	B	C	C	D	B	B	B	C
27	28	29	30	31	32	33	34	35	36	37	38	39

D	C	C	C	C	C	B	A	C	B	C	A	B
---	---	---	---	---	---	---	---	---	---	---	---	---

Write True or False, and correct the FALSE statements.

- 1) All users can assign negative nice values
- 2) A thread is an independent execution unit with its own memory space.
- 3) Threads allow different parts of a program to run concurrently in different processes.
- 4) The command ps is Useful for monitoring system performance and identifying resource-heavy tasks.
- 5) A process in the "Stopped" (T) state can be resumed using the SIGCONT signal.
- 6) Daemon programs are background processes that do not have a user interface.
- 7) Nice values range from -20 (lowest priority) to +19 (highest priority).
- 8) The ps command shows only kernel processes by default.
- 9) SIGCHLD is sent to the child when its parent terminates.
- 10) The scripts that the init process runs are located in /etc and called init scripts
- 11) All processes are in /etc

Answers

- 1) False
Only root can assign negative nice values
- 2) False
A thread is a lightweight subprocess that shares memory with other threads in the same process.
- 3) False
They allow different parts of a program to run concurrently within the same process.
- 4) False
The command Top is Useful for monitoring system performance and identifying resource-heavy tasks.
- 5) True
- 6) True
- 7) False
-20 is highest and +19 is lowest

8) False

ps shows user processes by default.

9) False

SIGCHLD is sent to the parent when a child terminates.

10) True

11) False

All processes are in /proc

4. Crontab and IP Addresses

Multiple Choice Questions (MCQ)

1. Which cron runs every day at 9 AM and 5 PM

- a) 0 9,17 * * *
- b) 0 9-17 * * *
- c) 0 9,5 * * *
- d) 0 9 * * * and 0 17 * * *

2. Which cron runs every Friday at 6:00 PM

- a) 0 18 * * 5
- b) 0 18 * * 6
- c) 0 6 * * 5
- d) 0 6 * * 6

3. Which cron runs on the first day of every month at midnight

- a) 0 0 1 * *
- b) 0 0 * * 1
- c) 0 10 * *
- d) 1 0 0 * *

4. Which statement best describes how cron decides to run a job

- a) The job runs only if all six fields match the current time
- b) The job runs when minute, hour, and month match, and either day of month or day of week matches
- c) The job runs only if day of month and day of week both match
- d) The job runs immediately after saving the crontab



5. Which command is used to edit a user's crontab

- a) crontab -l
- b) sudo edit crontab
- c) vim /etc/crontab
- d) crontab -e

6. Which cron expression runs a job every 15 minutes

- a) 0,15 * * * *
- b) */15 * * * *
- c) 15 * * * *
- d) */15 * * *

7. Which of the following cron jobs runs every Sunday at 12:00 AM

- a) 0 0 * * 7
- b) 0 12 * * 0
- c) 0 0 * * 0
- d) 12 0 * * 0

8. Which command displays the current IP address in Ubuntu

- a) ipconfig
- b) netplan show
- c) ip addr
- d) route

ifconfig is the networking command for Linux, while ipconfig is its equivalent for Windows.

9. What is the main difference between Static IP and Dynamic IP

- a) Static IP is faster
- b) Dynamic IP cannot access the internet
- c) Static IP does not change automatically
- d) Dynamic IP needs manual configuration

10. In a netplan file, which line disables DHCP

- a) dhcp: false
- b) dhcp4: off
- c) disable-dhcp: true



d) dhcp4: no

11. Which mistake is most likely to cause a netplan configuration failure

- a) Using Google DNS
- b) Wrong indentation in YAML
- c) Using nano instead of vim
- d) Restarting the system

12. Should /24 be added to the IP address in netplan?

- a) yes ,To define the gateway
- b) yes , To specify the subnet mask
- c) yes , To enable DNS
- d) No , need it

13. Which command safely tests netplan configuration before applying it

- a) netplan apply
- b) netplan test
- c) netplan try
- d) netplan check

14. Which editor is installed by default on Ubuntu

- a) vim
- b) nano
- c) emacs
- d) vi

15. What does this cron job do: 0 * * * * free -h >> /home/user/mem.log

- a) Logs memory usage every minute
- b) Logs memory usage every hour at minute 0
- c) Clears memory every hour
- d) Restarts the system hourly

16. Where are user crontab files stored

- a) /etc/crontab
- b) /var/spool/cron/crontabs
- c) /home/user/cron
- d) /usr/bin/cron

17. When will it runs: 0 5 1 * 1

- a) Only if the 1st of the month is a Monday
- b) Every 1st of the month at 5 o'clock
- c) Every Monday at 5 o'clock
- d) B and C together

18. When will it run : 30 23 * * *

- a) 11:30 AM
- b) 11:30 PM
- c) 12:30 AM (midnight)
- d) Every 23 minutes

19. What does the comma (,) mean in cron syntax

- a) Range of values
- b) Step values
- c) List of values
- d) Comment separator

20. What does 0-23/2 mean in the hour field

- a) Every other hour (0, 2, 4, ..., 22)
- b) Twice per hour
- c) Every 2 hours starting at hour 0
- d) A and C

21. Which timezone does cron use by default

- a) UTC
- b) The system's local timezone
- c) User's configured timezone
- d) GMT

22. What is the purpose of escaping % with \ in cron commands

- a) To enable special cron features
- b) To prevent cron from interpreting % as a newline
- c) To make the command run faster
- d) To allow comments in the command

23. Which configuration sets a dynamic IP (DHCP) in Netplan

a)

```
method: dhcp
```

b)

```
static: true  
ip: auto
```

c)

```
dhcp4: yes  
addresses: []
```

d)

```
dhcp4: no  
addresses: [192.168.1.100/24]
```

24. In a crontab expression, which field is represented by the 5th argument?

- a) Day of the month
- b) Month
- c) Day of the week
- d) Command to be executed

25. What is the meaning of */15 in the minute field?

- a) At minute 15 only
- b) Every 15 minutes starting from minute 1
- c) Every 15 minutes starting from 0
- d) At minute 0 and 15 only

26. Which command edits the root user's crontab?

- a) crontab -e
- b) sudo crontab -e
- c) crontab -u root -l
- d) su crontab -e

27. Which statement about md5sum is correct?

- a) It encrypts files
- b) It compresses files
- c) It generates a checksum to detect changes
- d) It hides file contents

Answers

1	2	3	4	5	6	7	8	9	10	11	12	13	14
A	A	A	B	D	B	C	C	C	D	B	B	C	B
15	16	17	18	19	20	21	22	23	24	25	26	27	
B	B	D	B	C	D	B	B	C	C	C	B	C	

Write True or False, and correct the FALSE statements

- 1) Cron jobs run immediately after saving
- 2) Netplan configuration files must end with .yaml
- 3) Wrong indentation can break a netplan configuration
- 4) Cron wakes up every second to check jobs.
- 5) "sudo netplan apply" permanently applies configuration
- 6) The command ip a is used to view the current IP address configuration of the machine.
- 7) The md5sum tool is used to encrypt sensitive files like /etc/passwd to ensure their secrecy
- 8) Commands are executed by Cron when all time arguments matches the current time
- 9) The Cron task syntax consists of 6 arguments separated by commas
- 10) We can use a forward slash with an asterisk (* /) to express a range of values

Answers

- 1) False
Cron waits until the scheduled time
- 2) True
- 3) True
- 4) False
Cron wakes up every minute.
- 5) True
- 6) True

7) False

md5sum is used for integrity checking (calculating a digital fingerprint/checksum) to detect changes, not for encryption or secrecy.

8) False

Commands are executed by Cron when the minute, hour and month fields match the current time, and when at least one of the day fields , either day of month, or day of week match the current time.

9) False

The Cron task syntax consists of 6 arguments separated by spaces

10) False

the asterisk with forward slash is used to express a step value

When will these commands be executed?

30 2 1 * 1 command

Answer:

When the day is Monday or the day is the 1st at 2:30 AM

0 0 * * 7 command

Answer:

Error, Day of week 7 is invalid. It must be in the range from 0 to 6

Write the cron Expression that excutes the command At 20 and 40 minutes past the hour, every 8 hours between July and December

Answer:

20,40 */8 * 7-12 *

5. Variables and Functions in Shell Script

Multiple Choice Questions (MCQ)

1. Which of the following best describes a shell script?

- a) A binary program compiled by the kernel
- b) A text file containing a sequence of shell commands
- c) A configuration file used only by system administrators
- d) A program written only in C language

2. What is the purpose of the following line? `#!/bin/bash`

- a) It comments out the first line
- b) It tells the kernel which interpreter to use
- c) It defines a global variable
- d) It gives execution permission

3. Which command gives execution permission to a script named test.sh?

- a) `chmod test.sh`
- b) `chmod 222 test.sh`
- c) `chmod +x test.sh`
- d) `./chmod test.sh`

4. Which of the following is a System Defined Variable (SDV)?

- a) `myVar`
- b) `username`
- c) `PATH`
- d) `RESULT`

5. Which statement about SDVs is correct?

- a) They must always be lowercase
- b) They are created only by users
- c) They are predefined by Linux and usually uppercase
- d) They cannot be printed

6. What is the output of: `echo $PWD`

- a) The previous working directory
- b) The current working directory
- c) The home directory



d) The shell path

7. What does \$0 represent?

- a) The first argument
- b) The process ID
- c) The current script name
- d) The exit status

8. What is the difference between \$* and \$@ when used inside double quotes?

- a) They behave identically
- b) \$* treats all arguments as one string, \$@ treats each as a separate string
- c) \$@ merges all arguments into one string
- d) \$* separates arguments into multiple strings

9. What is the value of \$# if the script is executed as: `./script.sh A B "C D"`

- a) 3
- b) 2
- c) 4
- d) 1

10. What does \$? store?

- a) PID of current process
- b) Number of arguments
- c) Exit status of the last command
- d) Script name

11. Which variable stores the PID of the current shell?

- a) \$!
- b) \$#
- c) \$?
- d) \$\$

12. Which command correctly performs integer arithmetic?

- a) `result = a + b`
- b) `result=$((a + b))`
- c) `result=$(a + b)`
- d) `((result = "a + b"))`

13. What happens if Bash tries to divide by zero?

- a) It returns 0
- b) It returns infinity
- c) It prints a warning and continues
- d) It throws a runtime error

14. Which tool is used for floating-point arithmetic in Bash?

- a) bc
- b) awk
- c) expr
- d) float

15. Which is a correct function definition in Bash?

- a) function = myFunc() {}
- b) myFunc () { echo Hello; }
- c) def myFunc() {}
- d) myFunc: echo Hello

16. What happens if a function is called before it is defined in a script?

- a) It works normally
- b) It executes later automatically
- c) It will cause a 'command not found' error.
- d) Bash ignores the function

17. Which keyword makes a variable local to a function?

- a) Private
- b) Static
- c) Hidden
- d) local

18. Which test operator checks if a file exists and is a regular file?

- a) -d
- b) -z
- c) -f
- d) -e

19. Which operator is used for numeric equality comparison?

- a) -eq
- b) ==
- c) =
- d) ===

20. Why is `[[ ... ]]` preferred over `[ ... ]` in modern Bash?

- a) It is faster
- b) It supports regex and is safer with spaces
- c) It is older
- d) It works only for numbers

21. Which operator is used within `[[ ]]` to compare a variable against a Regular Expression (regex)?

- a) ==
- b) -eq
- c) =~
- d) -z

22. What is the default shell for a new user created with 'useradd' in many Linux systems?

- a) /bin/bash
- b) /bin/sh
- c) /bin/ksh
- d) /usr/bin/zsh

23. What is the result of the arithmetic operation `$((5 / 2))` in a standard Bash script?

- a) 2.5
- b) Error
- c) 3
- d) 2

24. How do you pass arguments to the function "my_func"?

- a) `my_func(arg1, arg2)`
- b) `my_func --args=arg1,arg2`
- c) `my_func arg1 arg2`
- d) `call my_func arg1 arg2`

25. What is the result of using the ':' (colon) operator in a 'then' block?

- a) It prints the previous command's output.
- b) It terminates the script immediately.
- c) It acts as a 'no-op' (does nothing) but satisfies syntax requirements.
- d) It comments out the rest of the line.

26. What is the output of the script:

```
#!/bin/bash
USERNAME="USER"
echo $USERNAME
```

- a) USER
- b) The username of the logged in user
- c) Error message because the variable name "USERNAME" is a System defined variable

Answers:

1	2	3	4	5	6	7	8	9	10	11	12	13
B	B	C	C	C	B	C	B	A	C	D	B	D
14	15	16	17	18	19	20	21	22	23	24	25	26
A	B	C	D	C	A	B	C	B	D	C	C	A

Write True or False, and correct the FALSE statements.

- 1) A shell script must always end with .sh.
- 2) All variables in Bash are global unless declared with local.
- 3) \$@ and \$* behave the same even inside double quotes.
- 4) \$! returns the PID of the last background process.
- 5) If \$? is 0, the previous command failed.
- 6) read -p prints a message and takes user input at the same time.
- 7) Bash allows division by zero and returns 0 silently.
- 8) A function in Bash can accept arguments just like a script.
- 9) The variable HOME stores the current working directory.
- 10) [["\$a" =~ ^[0-9]+\$]] checks if a is a non-negative integer.



11) -z "\$var" returns true if the variable is not empty.

12) chmod +x script.sh executes the script.

Answers:

1) False

A script does not need .sh; it is only a convention, the system identifies the script type via the Shebang

2) TRUE

3) FALSE

Inside quotes, \$* joins arguments into one string, \$@ keeps them separate.

4) TRUE

5) FALSE

\$? = 0 means success, not failure.

6) TRUE

7) FALSE

Division by zero causes an error and stops execution

8) TRUE

9) False

HOME stores the user's home directory, not the current directory. (PWD stores current directory.)

10) TRUE

11) FALSE

-z "\$var" is true if the variable is empty.

12) FALSE

chmod +x only gives execution permissions. Execution is done by ./script.sh.

Write the Output of the Following Bash Scripts

Assume each script is saved in a file and executed correctly.

1. First script

```
#!/bin/bash
x = 5
y = 6
sum = ((x + y))
echo $sum
```

Answer:

The script has errors, Spaces around = are illegal in Bash assignments.

Corrected version:

```
#!/bin/bash
x=5
y=6
sum=$((x + y))
echo $sum
```

Output: 11

2. Second script

```
#!/bin/bash
if ["$a" -gt 5]
then
    echo "Greater"
```

Answer:

The script has errors:

- Missing spaces around the square brackets (In Bash, "[" and "]" are commands and must be separated by spaces.)
- Variable a is not defined
- Missing fi

Corrected version:

```
#!/bin/bash
a=7
if [ "$a" -gt 5 ]
then
    echo "Greater"
fi
```

Output: Greater

3. Third script, assume The user inputs 5 for 'val' when prompted

```
#!/bin/bash
x=10
y="x"

read -p "Enter a number: " val
result=$(( val + y ))
echo "The final result is:
$result"
```

Output: The final result is:

15

4. Fourth Script

```
#!/bin/bash
check() {
    if [ "$1" -gt 10 ]; then
        return 0
    else
        return 1
    fi
}

check 8
echo "Exit status: $?"

check 20
echo "Exit status: $?"
```

Output:

Exit status: 1

Exit status: 0

5. Fifth script

```
#!/bin/bash
a=5
b=2
result=$(a + b)
echo $result
```

Answer:

The Script has errors, it will give: (bash: a: command not found), because Bash attempts to execute *a* as a command.

Corrected version: (needs double parenthesis)

```
#!/bin/bash
a=5
b=2
result=$((a + b))
echo $result
```

Output: 7

6. Sixth script, assume the script is Executed as ./script.sh "I Love" OS

```
#!/bin/bash
echo "Using \${*}:"
for i in "${*}"; do
    echo "$i"
done

echo "Using \${@}:"
for i in "${@}"; do
    echo "$i"
done
```

Output:

Using \${*}:

[I Love OS]

Using \${@}:

[I Love]

[OS]

6. Loops and Case Statement

1. In a select loop, if a user enters a string (e.g., "abc") instead of a number, what happens:

```
select choice in Apple Banana Orange
do
    echo "You choice: $choice"
    break
done
```

- a) The script crashes
- b) The string is stored in the variable and matched against patterns
- c) The prompt repeats without executing any command
- d) The variable is set to null (Empty)
- e) C&D

2. In a case statement, if a variable matches multiple patterns, what is the behavior of the script

- a) All matching clauses are executed
- b) Only the first matching pattern's commands are executed
- c) The script throws a syntax error
- d) Only the last matching pattern is executed

3. Consider an until loop and a while loop with the same condition [-f "ready.txt"]. What is the fundamental difference in their execution

- a) There is no difference
- b) The while loop runs as long as the file exists; the until loop runs until the file is created
- c) The until loop checks the condition after executing the body, while checks it before
- d) The while loop requires a break statement to stop, whereas until does not

4. How does the select loop behave after executing a command for a valid choice

- a) It exits automatically
- b) It waits for 5 seconds and then exits
- c) It displays the menu and prompt again
- d) It clears the screen

5. What is the role of the PS3 variable in a select loop

- a) It sets the default value for the selection variable
- b) It sets the custom prompt text displayed to the user
- c) It limits the number of options in the menu
- d) It defines the path to the shell

6. If PS3 is not set by the programmer, what is the default prompt for select

- a) \$
- b) <
- c) #?

7. Predict the output of the following until loop if a file named test.txt already exists

```
until [ ! -f test.txt ]; do  
    echo "Inside"  
    break  
done
```

- a) Inside
- b) No output
- c) Infinite loop of "Inside"
- d) Error: file exists

8. How do you separate multiple patterns in a single case clause

- a) Using a comma(,)
- b) Using a semicolon(;)
- c) Using a vertical bar(|)
- d) Using the AND keyword

9. Which command is used to terminate a select loop from within its body

- a) done
- b) exit
- c) break
- d) B and C

10. The keyword used to close a case statement is

- a) endcase
- b) esac
- c) done
- d) fi

11. What is the exit status of a case statement if a pattern matches

- a) Always 0
- b) Always 1
- c) The exit status of the last command executed in the matching clause
- d) The exit status of the case keyword itself

12. What symbol is used to end each specific clause (pattern and its commands) in a case statement

- a) ;
- b) ::
- c) ;;
- d) |

13. In the until loop, when does the execution of commands inside the loop stop

- a) While the condition is false
- b) Once the condition becomes true
- c) When the control expression returns a non-zero status
- d) After exactly three iterations

14. What happens if no patterns match in a case statement and no default * pattern is provided?

- a) The script terminates with an error.
- b) The statement returns an exit status of 0 and moves to the next command.
- c) The script enters an infinite loop.
- d) The first pattern is executed by default.

15. When using the select loop, what does the variable defined in the loop header (e.g., select choice in ...) store?

- a) The number of the choice entered by the user.
- b) The literal string value of the selected option.
- c) The total count of options available.
- d) The exit status of the last command executed.

- The numeric choice entered by the user is actually stored in the built-in REPLY variable.
- If the user enters a string instead of the choice number, the string is stored in the REPLY variable and the variable in the select statement is set to null (empty)

16. What is the exit status of a case statement if no pattern matches?

- a) 1
- b) -1
- c) 0
- d) Same as the last command

17. Which wildcard matches exactly one character in case patterns?

- a) *
- b) ?
- c> []
- d) |

18. In this command, what does -f2 mean?

cut -d':' -f2

- a) Second file
- b) Second field
- c) File size 2
- d) Force mode

19. What happens after executing exec bash inside a script?

- a) A child shell is created
- b) The script pauses
- c) The script is replaced by a new Bash process
- d) The script continues normally

Answers:

1	2	3	4	5	6	7	8	9	10
D	B	B	C	B	C	A	C	C	B
11	12	13	14	15	16	17	18	19	
C	C	B	B	B	C	B	B	C	

True or False

- 1) The default pattern in case is written as ?)
- 2) -d in [-d path] checks if a file exists.
- 3) The for loop is primarily used to iterate over a series of words within a string or a list of items.

Answers

- 1) False
default pattern is *)
- 2) False
-d checks for directory
- 3) True

What is the output?

1.

```
#!/bin/bash
for i in 1 2 3;
do
    echo -n $i
done
```

Output:

123

(the shebang #/bin/bash is invalid (missing !), so the script is only guaranteed to work as shown if it is executed explicitly by Bash (e.g., bash script.sh). Otherwise, the interpreter is undefined.)

2.

```
VAR="apple"
case $VAR in
  app*) echo "Matched 1" ;;
  apple) echo "Matched 2" ;;
  *) echo "No Match" ;;
esac
```

Output:

Matched 1

3.

```
i=0
while ((i<3)); do
  case $i in
    1) ((i+=2)) ;;
    *) ((i++)) ;;
  esac
done
echo $i
```

Output:

3

4. `cat /etc/passwd | cut -d: -f1 | tail -n 1` (Assume the last user created on the system was 'testuser')

Output:

testuser

Explanation: This command pipes the user list, extracts the first field (username), and displays the very last entry

5. `sudo useradd -m -s /bin/bash exam_user`
`grep "exam_user" /etc/passwd | cut -d: -f7`

Output:

/bin/bash

Explanation: The cut command extracts the 7th field of the /etc/passwd entry, which is the default shell.

6.

```
#!/bin/bash
attempt=2
if [ $attempt < 3 ]; then
    echo "$attempt is less than 3"
fi
```

Output:

Error, < is treated as an input redirection operator, and Bash attempts to read from a file named 3

Corrected version:

```
#!/bin/bash
attempt=2
if [ $attempt -lt 3 ]; then
    echo "$attempt is less than 3"
fi
```

7.

```
#!/bin/bash
for i in 1 2 3
    echo "Value: $i"
done
```

Output:

ERROR

Explanation: The for loop is missing the mandatory do keyword.

Corrected version:

```
for i in 1 2 3; do
    echo "Value: $i"
done
```

8.

```
#!/bin/bash
case "Apple" in
    [a-z]*) echo "Starts with lowercase"
    ;;
    [A-Z]*) echo "Starts with uppercase"
    ;;
    *) echo "Unknown" ;;
esac
```

Output:

Starts with uppercase

Explanation: "Apple" starts with 'A', which matches the pattern [A-Z]* (any string starting with an uppercase letter)

9.

```
x="file1"
case $x in
    file?) echo "One digit file" ;;
    file*) echo "Any file" ;;
esac
```

Output:

One digit file

Explanation: file? matches before file*

7- Arrays

1. What is an array

- a) A Bash command
- b) A variable that stores a single value
- c) A data structure that stores multiple values under one name
- d) A type of file

2. Which command correctly declares an indexed array

- a) declare -A arr
- b) arr=(3 2 1)
- c) arr=[3 2 1]
- d) array arr

3. How do you append an element to an indexed array

- a) arr.add(1)
- b) arr=("x")
- c) arr+=("x")
- d) arr++

4. What is the main difference between `${arr[@]}` and `${!arr[@]}`

- a) First prints indices, second prints values
- b) First prints values, second prints keys
- c) Both print the same
- d) First is for associative arrays only

5. Which command prints the number of elements in an array

- a) `${arr[*]}`
- b) `${arr[@]}`
- c) `${#arr[@]}`
- d) `${!arr[@]}`

6. Which command is used to delete element from array

- a) delete a[1]
- b) unset a[1]
- c) rm a[1]
- d) remove a[1]

7. Which command lists systemd service unit files

- a) ps aux
- b) systemd list-unit-files
- c) systemctl list-unit-files
- d) service --status-all

8. What is the purpose of the last line in this code

```
#!/bin/bash
while read -r user; do
    echo "User: $user"
done < <(cut -d: -f1 /etc/passwd)
```

- a) It writes user name to /etc/passwd
- b) It reads input from the keyboard
- c) It passes the output of the command to the while loop
- d) It executes the loop only once

9. What does ls -t do

- a) Sort by name
- b) Sort by size
- c) Show hidden files
- d) Sort by modification time

10. What does --no-legend do in systemctl

- a) Hides unit names
- b) Removes header lines
- c) Disables services
- d) Hides inactive services

11. Which tool is commonly used in Bash to send messages to Telegram

- a) send -te
- b) curl
- c) ssh
- d) scp

12. What does the (-X POST) with the previous command mean

- a) Runs in background
- b) Sends HTTP POST request
- c) Deletes Telegram messages
- d) Saves output to a file

13. What Parse Mode can be used for the Telegram message format

- a) Markdown
- b) HTML
- c) JSON
- d) All of the above

14. What basic data is required to set up the Telegram Bot in the script

- a) Machine username and password
- b) BOT_TOKEN and CHAT_ID
- c) Only the server's IP address
- d) A link to Google

15. In the context of the Telegram Bot API, which curl flag is used to send data via a POST request?

- a) -G
- b) -X POST
- c) -s
- d) -o /dev/null

16. In an associative array: `user[role]="DevOps"`
What happens if key role already exists?

- a) Bash throws an error
- b) A new element is appended
- c) The value is overwritten
- d) The array is cleared

17. Which service state means it can never be started, even manually?

- a) disabled
- b) static
- c) masked
- d) enabled-runtime

Answers

1	2	3	4	5	6	7	8	9
C	B	C	B	C	B	C	C	D
10	11	12	13	14	15	16	17	
B	B	B	D	B	B	C	C	

True or False

- 1) By default, bash indexed arrays start their numbering from 1
- 2) The command `ls -t` is used in log rotation scripts because it sorts files by name in alphabetical order
- 3) The command `unset arr[2]` deletes the entire `arr` array from memory
- 4) `Curl -s -o /dev/null` hides both progress and server response.

ANSWERS:

- 1) **False.**
Bash indexed arrays start numbering from 0
- 2) **False.**
`Ls -t` sorts files by modification time (newest first), which is essential for identifying old logs for rotation
- 3) **False.**
`Unset arr[2]` only deletes the element at index 2.
To delete the whole array, use `unset arr`.
- 4) **True**

What is the output?

1.

```
#!/bin/bash
my_list = ("A" "B" "C")
echo ${my_list[0]}
```

Output:

Error, there must be **no spaces** around the assignment operator (=). It should be my_list=("a" "b" "c")

2.

```
#!/bin/bash
p=(1,2,3)
echo ${p[0]}
echo ${p[1]}
echo ${p[2]}
```

Output:

1,2,3 with two newlines

Explanation: when values are separated with commas they considered one value.

8. Docker

Multiple Choice Questions (MCQ)

1. What is the default storage path for Docker Volumes on the host system?

- a) /var/lib/docker/containers
- b) /usr/local/docker/volumes
- c) /var/lib/docker/volumes
- d) /opt/docker/volumes

2. Which of the following is listed as a Type 1 Hypervisor example?

- a) VMware Workstation
- b) Oracle VirtualBox
- c) VMware Fusion
- d) Oracle VM Server

3. Containers achieve isolation using which TWO specific Linux kernel features?

- a) SELinux and AppArmor
- b) namespaces and cgroups
- c) iptables and netfilter
- d) seccomp and capabilities

4. In the context of containerization, what is shared between the host and the containers?

- a) The entire Guest Operating System.
- b) Hardware-level abstraction only.
- c) The Host OS Kernel.
- d) The physical BIOS settings only.

5. What is the startup time listed for Virtual Machines?

- a) Seconds
- b) Hours
- c) Minutes
- d) Milliseconds

6. Which network driver does explicitly state is the DEFAULT for standalone containers?

- a) Overlay
- b) Host
- c) Bridge
- d) Macvlan

7. tmpfs data is stored where?

- a) In Docker volumes directory
- b) In host's RAM (or swap when full)
- c) In container's writable layer
- d) In /tmp directory on host

8. Which statement about Docker Images is correct?

- a) Images are executable and run processes
- b) Images are read-only templates with instructions
- c) Images are stored in RAM for faster access
- d) Images are automatically compressed

9. When you rebuild a Docker image after modifying the Dockerfile:

- a) The entire image is rebuilt from scratch
- b) Only the final layer is rebuilt
- c) All layers are rebuilt but cached
- d) Only the changed layers are rebuilt

10. Microsoft Hyper-V is categorized as:

- a) Type 2 Hypervisor
- b) Container Runtime
- c) Type 1 Hypervisor
- d) Both Type 1 and Type 2

11. Docker Volumes are managed via:

- a) Host operating system file manager
- b) Docker CLI (command line)
- c) Docker Hub interface
- d) Manual configuration files

12. Which best use case is listed for Containers

- a) Enterprise systems requiring multiple OS
- b) High security compliance environments
- c) Microservices applications
- d) Running different operating systems

13. What happens when a container is deleted and data was stored in its writable layer?

- a) Data is automatically backed up
- b) Data is moved to a volume
- c) Data is permanently lost
- d) Data is transferred to the image

14. Overlay networks are primarily used with:

- a) Standalone containers on one host
- b) Docker Swarm for multi-host communication
- c) Kubernetes clusters only
- d) Single-container applications

15. What does Macvlan network assign to each container?

- a) A unique IP address only
- b) A unique container name
- c) A unique MAC address
- d) A unique port number

16. tmpfs mounts can be shared between containers ONLY on which OS?

- a) Windows
- b) macOS
- c) Any Unix system
- d) Linux

17. Which hypervisor type runs directly on physical hardware?

- a) Type 2 (Hosted)
- b) Type 3 (Container-based)
- c) Type 1 (Bare Metal)
- d) Type 0 (Firmware-based)

18. Docker Hub is described as:

- a) A public registry that is open source and default
- b) A local storage system for images
- c) A localized database for managing virtual machines
- d) A private registry requiring payment

19. When you run `docker pull ubuntu`, where is the image downloaded from by default?

- a) Docker Daemon
- b) Docker Client
- c) Docker Hub
- d) Local machine storage

20. When running a container from a non-existent local image, Docker will:

- a) Throw an error and stop
- b) Create an empty container
- c) Search in the configured registry and pull it automatically
- d) Ask user for manual confirmation

21. Which Docker volume mount flag is described as "newer with more flexible options"?

- a) `--tmpfs`
- b) `--volume`
- c) `--storage`
- d) `--mount`

22. A company wants to run two different applications on the same server. App A requires a Linux kernel version 5.15, while App B requires a Windows-specific kernel. Which technology must be used?

- a) Docker Containers
- b) Virtual Machines (VMs)
- c) Docker Swarm
- d) Type 2 Hypervisor only

23. You have multiple containers running on different physical servers (Host A and Host B). You need these containers to communicate with each other as if they were on the same local network. Which driver is required?

- a) Bridge Network
- b) Host Network
- c) None Network



d) Overlay Network

24. A developer uses a container to process sensitive bank transactions. To comply with security laws, the data must never reside on a non-volatile disk (Hard Drive). Which mount is best?

- a) `-v /my/data:/app/data` (Bind Mount)
- b) `--mount type=volume` (Docker Volume)
- c) `--tmpfs /app/data` (Tmpfs Mount)
- d) A Writable Layer change

25. If the dockerd process (Daemon) crashes, what happens to the user's ability to manage containers via the docker command-line interface (CLI)?

- a) It continues working because the CLI is independent.
- b) It stops working because the CLI cannot communicate with the "brain" that manages images and containers.
- c) The CLI automatically becomes the new Daemon.
- d) Only the network is affected, but containers can still be started.

26. Which of the following is a primary advantage of Virtual Machines (VMs) over Containers?

- a) Faster startup times (measured in seconds).
- b) Stronger, hardware-level isolation for high-security environments.
- c) Minimal resource overhead and disk space usage.
- d) Easier integration with CI/CD pipelines.

27. What is the primary role of the Docker Daemon (dockerd)?

- a) To provide a Command Line Interface (CLI) for users to type commands.
- b) To act as a registry for storing and sharing public images.
- c) To manage Docker objects like images, containers, networks, and volumes.
- d) To serve as the primary GUI for managing containers on Windows.

28. Which of the following best explains why containers start faster than virtual machines?

- a) Containers do not use storage drivers
- b) Containers share the host OS kernel instead of booting a full OS
- c) Containers use Type 1 hypervisors
- d) Containers do not require CPU scheduling

29. Which statement about isolation is most accurate?

- a) Containers provide stronger isolation than VMs
- b) VMs isolate at hardware level, containers at process level
- c) Containers and VMs both isolate using hypervisors
- d) Containers isolate using BIOS virtualization

30. When a container is deleted, which data is lost?

- a) Data in Docker volumes
- b) Data in bind mounts
- c) Data in tmpfs
- d) Data in writable layer not backed by persistent storage

31. Which pair correctly matches component to role?

- a) Docker Daemon → Manages Docker objects
- b) Docker Registry → Executes containers
- c) Docker Client → Stores images
- d) Dockerfile → Provides runtime kernel

Answers

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
C	D	B	C	C	C	B	B	D	C	B	C	C	B	C	D
17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	
C	A	C	C	D	B	D	C	B	B	C	B	B	D	A	

TRUE OR FALSE

- 1) A Type 2 Hypervisor is more performant than a Type 1 Hypervisor because it benefits from the host OS's drivers.
- 2) Docker Images are "Read-Only" templates, while Containers are "Runnable Instances" of those images.
- 3) By default, data stored inside a container's writable layer is persistent and will remain available even after the container is deleted.

- 4) The "Bridge" network is the default network driver for standalone containers and allows containers on the same host to communicate.
- 5) Each virtual machine must include its own kernel, while containers share the host kernel.
- 6) Bridge networking removes network isolation between host and container.
- 7) A container is defined only by its image and cannot be customized at runtime.
- 8) Docker volumes remain even after the container that used them is deleted.
- 9) Hypervisors provide stronger isolation than containers because they virtualize hardware.
- 10) Bind mounts are fully managed by Docker and independent of host filesystem structure.
- 11) Docker can communicate with other daemons to manage docker services

Answers

- 1) False.

Type 1 Hypervisors are more performant than Type 2 because they run directly on the hardware (bare metal) without the overhead of a host operating system

- 2) True

- 3) False.

Data in a container's writable layer is non-persistent; it is lost once the container is deleted. Persistent storage requires the use of Volumes or Bind Mounts.

- 4) True

- 5) True

- 6) False

Host network removes isolation; Bridge provides isolated networking.

- 7) False

A container is defined by its image and runtime configuration options.

- 8) True

- 9) True

- 10) False

Bind mounts depend directly on the host filesystem and are not fully managed by Docker.

- 11) True

	Virtualization	Containerization
Definition	Runs full operating systems as “guest” machines on a physical host via a hypervisor.	Packages applications with dependencies into containers that share the host OS kernel
Execution Layer	Hypervisor sits on hardware (Type 1) or on top of host OS (Type 2)	Container runtime (e.g., Docker Engine) runs on top of the host OS
Operating System	Each VM requires its own complete OS (Windows, Linux, etc.)	Containers share the host OS kernel; no separate OS needed
Isolation	Strong hardware-level isolation; suitable for high-security environments.	Process-level isolation using namespaces and cgroups; lighter but less strict
Resource Consumption	High (RAM, CPU, storage) due to multiple OS instances	Low; containers only use resources needed for the application.
Startup Time:	Slower (minutes)	Fast (seconds).
Performance	Near-native performance but with overhead	Lightweight, near-native performance with minimal overhead
Portability	Limited to hypervisor and OS compatibility	Highly portable across environments (Docker, Kubernetes)
Management	Complex, requires patching, updating, and managing multiple OS instances.	Simplified, easy updates, redeployment, CI/CD integration.
Security	Strong isolation; each VM is fully separated.	Good but depends on host OS security; needs hardening.
Best Use Cases:	Enterprise workloads, compliance-heavy environments, multi-OS requirements	Microservices, cloud-native apps, DevOps pipelines, rapid scaling
Examples	VMware ESXi, Microsoft Hyper-V, Oracle VM, Citrix Hypervisor.	Docker, Kubernetes, Podman, OpenShift



RBC^s